

A sonic black hole of a rectangular cross-section

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ABSTRACT

In this study, a rectangular cross-section sonic black hole (SBH) is explored and a one-dimensional model equation for the acoustic field inside it is presented. The model equation's applicability is verified through a comparison with the corresponding two-dimensional Helmholtz equation. A general analytical solution, which takes into account the SBH's quadratically varying internal cross-section, is derived using linearly independent confluent Heun functions. To investigate the acoustic wave behavior in the SBH, the wave splitting method is utilized to obtain forward and backward acoustic pressure waves, and the reflection coefficient is calculated. To confirm the validity of the analytical solution, a comparison is made with numerical results obtained from the Riccati equation. Within the context of this study, a detailed discussion is provided regarding the applicability of the analytical solution.

1. Introduction

Acoustics is a field that addresses various challenges related to the control and management of sound and vibration, including the suppression of unwanted noise or vibration. One promising tool in this regard is the use of acoustic black holes (ABHs), which have attracted the interest of many experts in recent years. In general, an ABH is a structure that significantly slows down both the phase and group velocities of acoustic waves, ideally to zero at the end of the structure. This causes incident waves not to reach the end of the ABH in finite time, hence the term “black hole” borrowed from astrophysics to describe these structures. The term ABH is used in the context of both sound and elastic waves, but most published papers have focused on elastic waves. However, this paper only addresses ABHs related to sound waves, which will be referred to as sonic black holes (SBHs) to avoid confusion. For those interested in ABHs related to elastic waves, the review paper by Pelat [1] provides a wealth of knowledge.

The concept of SBHs is based on the gradual reduction of the internal cross-section of a waveguide, while the admittance of the wall varies along the waveguide's longitudinal axis. The initial idea for SBHs was published in the paper [2], which proposed an SBH with a circular cross-section and a set of rings with a variable inner radius providing the required wall admittance. Subsequent papers such as [3–6] built upon this idea and explored various aspects of SBHs such as their discrete implementation, acoustic losses, and the role of critical and ring frequencies.

The authors of the paper [3] used the transfer matrix method to solve a one-dimensional model equation for SBHs and verified the results using analytical solutions for linearly and quadratically decreasing inner radius of the set of rings. In the paper [4], it was shown that the discrete SBH could be interpreted as a practical realization of a metafluid whose density increases as it approaches

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the waveguide end wall. The authors of [5] investigated open-end and closed-end SBHs and compared measured results with those obtained by solving the one-dimensional model equation using the transfer matrix method.

In contrast to the previous works, the authors of [6] developed a mathematical model based on the linearized Navier-Stokes equations that captures thermoviscous losses in the acoustic boundary layer along the fluid-solid interface, especially in the cavities between the rings.

The authors of [7] addressed the mechanism of losses in SBHs and concluded that the primary source of damping responsible for low reflection coefficients at higher frequencies is local absorption effects at the outer surface of the cylinder. In the paper [8], the authors analyzed the spectra of the acoustic power, radiation efficiency, and far-field acoustic pressure of SBHs and reported the role of critical and ring frequencies.

The absorption performance of SBHs of circular cross-section is studied in detail by the authors in the paper [9]. The study compares their numerical results with experimental data obtained using COMSOL. It is demonstrated that lightweight annular plate vibrations significantly affect the absorber performance over the entire frequency range, highlighting the need for a comprehensive understanding of the behavior of SBHs in this regard.

The absorption performance of SBHs is not limited to the aforementioned set of rings. For instance, in the paper [10], the authors designed and measured SBHs based on two other structures. The first structure consists of a series of rigid discs fixed on a rod and placed inside the tube, with the diameters of the discs gradually increasing according to a parabolic law. The second one is a tube with mass layers placed inside, where the effective density of the medium increases, and the concentration of these layers grows towards the end of the tube. Furthermore, in the paper [11], the authors proposed and numerically and experimentally investigated a sound-absorbing device that combines SBHs and microporated panels to achieve near-perfect sound absorption in the subwavelength thickness region, particularly excelling at low frequencies. Another new use of SBHs is described in [12], where the authors studied the sound wave propagation and attenuation characteristics in a new acoustic device constructed by periodically embedding SBHs into an air-filled waveguide. It is worth noting that other structures, such as those described in [13], have shown similar effects to SBHs. In this paper, the authors reported perfect acoustic absorption over a broad frequency band in deep-subwavelength thickness panels using the rainbow-trapping effect.

The papers mentioned above mainly focus on SBHs with axisymmetric cross-sections. In this paper, our aim was to design an SBH with a rectangular cross-section, where acoustic resonances in the slits between the ribs inside the SBH can occur at higher frequencies. This type of structure can be utilized for rectangular waveguides or panels. To achieve this, we derived an appropriate one-dimensional model equation and analytical solution for a specific shape of the cross-sectional function and the possibility of adjusting the length of the slits between the ribs. This work serves as a preliminary investigation, and more studies are needed to address further details, especially regarding the implementation of a more complex acoustic loss mechanism. Therefore, this paper is primarily theoretical in nature.

This paper is structured as follows. In Sec. 2, we present a one-dimensional model equation to describe acoustic waves in the SBH and provide the geometrical dimensions of the structure. In Sec. 3, we derive the corresponding Riccati equation for the reflection coefficient based on the model equation. Next, in Sec. 4, we present the analytical solution of the model equation for the quadratic cross-sectional function. We then perform a separation of the total acoustic field into forward and backward waves using the wave splitting method in Sec. 5 to analyze the behavior of acoustic waves inside the SBH. In Sec. 6, we validate the found analytical solution of the model equation by comparing it with the numerical solution of the Riccati equation, and we show that the proposed structure behaves as a SBH. The results are discussed in Sec. 7. Finally, we provide our conclusions in Sec. 8. In Appendix, we include a derivation of the model equation and features concerning the confluent Heun equation.

2. Model equation

We consider a harmonic sound wave incident on a rectangular sound-absorbing structure, whose geometric specification is shown in Fig. 1. As described in Section 6, this structure demonstrates the features of a sonic black hole, which we will henceforth refer to as such in this paper. The walls of the SBH are assumed to be perfectly rigid, and the structure is fitted with a set of rigid ribs whose width is much smaller than the distance w between them. Fig. 2 shows that the variable cross-section is described by the cross-sectional function $a_0(x)$ and the rib length $\ell(x)$ is controlled by the wall function $q(x)$. We assume only the plane mode, i.e., that the frequency f of the incident harmonic wave is lower than the cut-off frequency f_{off} of the waveguide, given by:

$$f_{\text{off}} = \frac{c_0}{4A_0} \quad \text{and} \quad b \leq 2A_0, \quad (1)$$

where c_0 is the sound velocity in gas and the meaning of A_0 , b can be seen in Fig. 1.

The model equation describing acoustic field inside the SBH is derived in Appendix A to this article and takes the following form

$$\frac{d^2 P(x)}{dx^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{dP(x)}{dx} + \left(\frac{j\omega\rho_0}{a_0(x)} Y_w(x) + \hat{k}_0^2 \right) P(x) = 0, \quad (2)$$

where

$$Y_w(x, \omega) = -\frac{j\hat{k}_0}{\omega\rho_0} \tan[\hat{k}_0\ell(x)] = -\frac{j\hat{k}_0}{\omega\rho_0} \tan[\hat{k}_0(q(x) - a_0(x))] \quad (3)$$

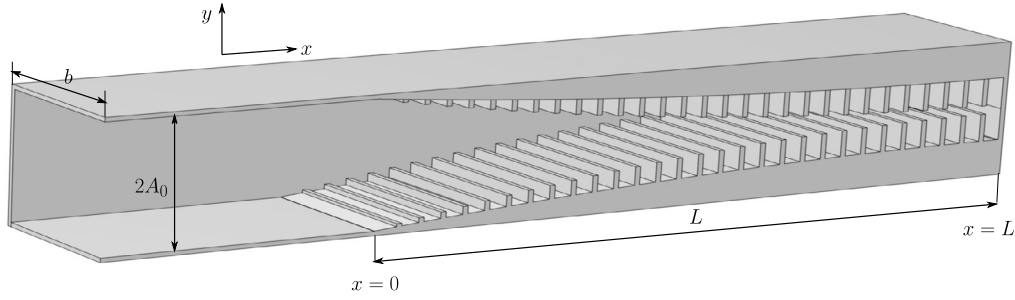


Fig. 1. Geometric specification of the considered SBH of rectangular cross-section.

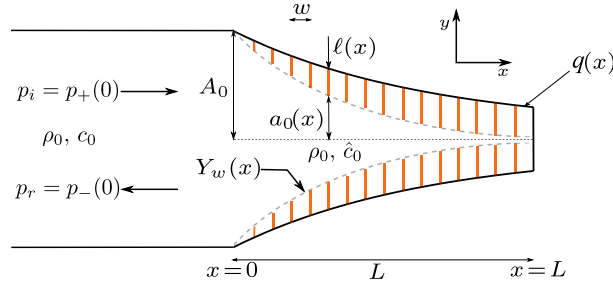


Fig. 2. Schematic capture of the considered SBH.

is the wall admittance. Here x is spatial coordinate in the direction of the SBH axis, t is time, $j = \sqrt{-1}$ is the imaginary unit, $\omega = 2\pi f$ is the angular frequency, $P(x)$ is the complex amplitude of acoustic pressure (assuming an implicit time dependence $\exp(-j\omega t)$), ρ_0 is the ambient (without sound) gas density, $\hat{k}_0 = \omega/\hat{c}_0$ is the complex wavenumber through which thermoviscous losses are included, where $\hat{c}_0 = c_0(1 - ja)$ is the complex propagation velocity (a is an ad-hoc parameter that accounts for thermoviscous losses). In several papers, such as [2,3], a value of $a = 0.05$ is used.

3. Riccati equation for the acoustic impedance and admittance

If we introduce small energy losses into the model equation (2) by adding an imaginary part to the sound velocity (see e.g., [2,3,5,6]), the resulting model equation can be used to calculate the reflection coefficient $R(x, \omega)$ as a function of spatial coordinate x and angular frequency ω . However, since we do not know the exact analytical solution of the model equation for a generally chosen cross-sectional functions $a_0(x)$ and $q(x)$, it is necessary to solve this problem numerically.

To accomplish this, we can convert Eq. (2) into the Riccati equation, which is a first-order nonlinear differential equation that is relatively easy to solve numerically.

The acoustic impedance can be expressed based on the linearized Euler equation for harmonic plane waves as

$$Z(x, \omega) = \frac{P(x)}{V_x(x)} = \frac{j\omega\rho_0 P(x)}{\frac{dP(x)}{dx}} \Rightarrow \frac{dP(x)}{dx} = \frac{j\omega\rho_0 P(x)}{Z(x, \omega)}, \quad (4)$$

where $V_x(x)$ is the complex amplitude of the component of the acoustic velocity in the x -axis direction.

By employing these relationships, the model equation (2) can be rewritten into the form

$$\frac{dZ(x, \omega)}{dx} = j\omega\rho_0 - \left(\frac{j\omega}{\rho_0 \hat{c}_0^2} - \frac{Y_w(x, \omega)}{a_0(x)} \right) Z^2(x, \omega) + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} Z(x, \omega), \quad (5)$$

which is the Riccati equation for the acoustic impedance. We can write for the acoustic impedance the following relation (see e.g., [14])

$$Z(x, \omega) = \rho_0 \hat{c}_0 \frac{1 + R(x, \omega)}{1 - R(x, \omega)}. \quad (6)$$

As we assume that the SBH is terminated by a perfectly rigid wall, it is preferable to rephrase the Riccati equation in terms of the acoustic admittance $Y(x, \omega) = 1/Z(x, \omega)$ for convenience:

$$\frac{dY(x, \omega)}{dx} = -j\omega\rho_0 Y^2(x, \omega) - \frac{1}{a_0(x)} \frac{da_0(x)}{dx} Y(x, \omega) - \frac{Y_w(x, \omega)}{a_0(x)} + \frac{j\omega}{\rho_0 \hat{c}_0^2}. \quad (7)$$

This equation is then solved numerically based on the initial condition for $Y(x = L, \omega)$.

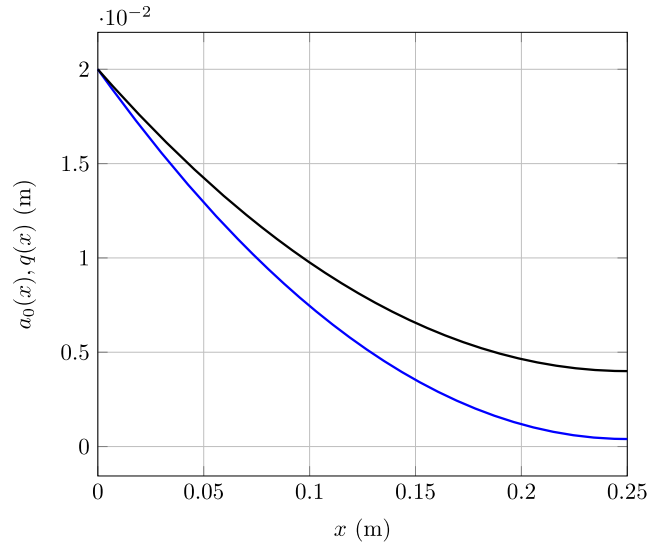


Fig. 3. Cross-sectional function $a_0(x)$ - black line ($A_0 = 0.02$ m, $a_{00} = 0.02$), the wall function $q(x)$ - blue line ($q_0 = 0$, $q_{00} = 0.2$).

By solving the Riccati equation, we can use the equation (7) to calculate the reflection coefficient as

$$R(x, \omega) = \frac{Z(x, \omega) - \rho_0 c_0}{Z(x, \omega) + \rho_0 c_0} = \frac{1 - \rho_0 c_0 Y(x, \omega)}{1 + \rho_0 c_0 Y(x, \omega)}. \quad (8)$$

The reflection coefficient $R(x = x_0, \omega)$, with $x_0 \in [0, L]$, characterizes the local reflection from an inhomogeneous medium. Specifically, in this medium, the region $[0, x_0]$ is replaced by a homogeneous medium with acoustic impedance $Z_0 = \rho_0 c_0$ in a waveguide with a rectangular cross-section of $S(x = x_0) = 2a_0(x_0)b$. This is a well-established concept in the literature, as discussed in [14,15].

4. Analytical solution of the model equation for the quadratic cross-sectional function

To better understand the behavior of the acoustic field inside the SBH, it is essential to have access to analytical solutions of the model equation (2). Although analytical solutions for linearly or quadratically decreasing inner radius of SBHs with circular cross-section have been published, as reported in [2,3], they are only applicable for low frequencies. Additionally, these solutions are not suitable for describing the acoustic field throughout the entire SBH due to the presence of singularities.

In order to avoid singularities in the model equation (2) and accurately describe the acoustic field inside the entire SBH, we propose using the following form of the cross-sectional function:

$$a_0(x) = A_0 \left(a_{00} + (1 - a_{00}) \left(\frac{x}{L} - 1 \right)^2 \right) = A_0 a_{00} \left(1 - a_2 \left(\frac{x}{L} - 1 \right)^2 \right), \quad a_{00} \leq 1, \quad (9)$$

where $a_2 = (a_{00} - 1)/a_{00}$. The parameter a_{00} is used to control the size of the SBH cross section at $x = L$.

We assume the wall function $q(x)$ in the same form like the cross-sectional function, i.e.

$$q(x) = A_0 \left(q_{00} + (1 - q_{00}) \left(\frac{x}{L} - 1 \right)^2 \right) + q_0 = A_0 q_{00} \left(1 - q_2 \left(\frac{x}{L} - 1 \right)^2 \right) + q_0, \quad a_{00} \leq q_{00}, \quad q_0 \geq 0, \quad (10)$$

where $q_2 = (q_{00} - 1)/q_{00}$. Examples of these two functions are shown in Fig. 3.

If $q(x) = a_0(x)$ and $q_0 = 0$, the model equation (2) becomes the classical Webster equation.

In order to solve Eq. (2) without being limited to very low frequencies, we can approximate the tangent function appearing in the relation for wall admittance (3) by a truncated Taylor series

$$\tan(\hat{k}_0 \ell(x)) \approx \hat{k}_0 \ell(x) \left(1 + \frac{1}{3} \hat{k}_0^2 \ell(x)^2 + \frac{2}{15} \hat{k}_0^4 \ell(x)^4 \right) \approx \hat{k}_0 \ell(x) (1 + F(\omega)), \quad (11)$$

where $\ell(x) = q(x) - a_0(x) = A_0(a_{00} - q_{00}) - (a_2 - q_2) \left(\frac{x}{L} - 1 \right)^2$ and

$$F(\omega) = \left(\frac{1}{3} A_0^2 (q_{00} - a_{00})^2 \hat{k}_0^2 + \frac{2}{15} A_0^4 (q_{00} - a_{00})^4 \hat{k}_0^4 \right). \quad (12)$$

This approximation represents a modification of the Taylor series expansion of the tangent function, which is made in order to facilitate finding an analytical solution to the model equation (2). Using the approximation (11), we can rewrite the wall admittance (3) as

$$Y_w(x, \omega) \approx -\frac{j\hat{k}_0}{\rho_0 \hat{c}_0} \ell(x)(1 + F(\omega)). \quad (13)$$

In previous papers that dealt with analytical solutions of the model equation for axisymmetric SBHs (see [2,3]), a similar approach was used, where $F(\omega)$ was set equal to zero. This approach results in a simplified approximation of the tangent function, which is given only by its argument. However, it's important to note that this simplified less accurate approximation ($F(\omega) = 0$) is only appropriate for very low frequencies. We will discuss this in more detail later.

If we consider the simplified approximation and $q(x) = A_0$ ($q_0 = 0$, $q_{00} = 1$), we obtain the model equation (2) in the following form:

$$\frac{d^2 P(x)}{dx^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{dP(x)}{dx} + \frac{\hat{k}_0^2 A_0}{a_0(x)} P(x) = 0. \quad (14)$$

This form of the model equation is similar to the model equation used for the SBH of the circular cross-section, see e.g., [2,3,5,6].

If we employ the wall admittance given by the more accurate formula (13) then we can write the model equation (2) as

$$\frac{d^2 P(x)}{dx^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{dP(x)}{dx} + \hat{k}_0^2 \frac{q(x) + (q(x) - a_0(x))F(\omega)}{a_0(x)} P(x) = 0. \quad (15)$$

To solve Eq. (15) we introduce the following new variable

$$\xi = a_2 \left(\frac{x}{L} - 1 \right)^2. \quad (16)$$

Using this variable, we rewrite Eqs. (9) and (10) as

$$a_0(\xi) = A_0 a_{00}(1 - \xi), \quad q(\xi) = A_0 q_{00} \left(1 - \frac{q_2}{a_2} \xi \right). \quad (17)$$

To rewrite Eq. (15) for the new variable ξ , we transform the derivatives

$$\frac{d}{dx} = \frac{2}{L} \sqrt{a_2 \xi} \frac{d}{d\xi}, \quad \frac{d^2}{dx^2} = \frac{4a_2 \xi}{L^2} \frac{d^2}{d\xi^2} + \frac{2a_2}{L^2} \frac{d}{d\xi}. \quad (18)$$

Employing the new variable (16), the expressions (17) and the transformation of derivatives (18), we can rewrite Eq. (15) into the following form:

$$\frac{d^2 P(\xi)}{d\xi^2} + \left(\frac{1}{2\xi} + \frac{1}{\xi - 1} \right) \frac{dP(\xi)}{d\xi} + \frac{\nu_c + \mu_c \xi}{\xi(\xi - 1)} P(\xi) = 0, \quad (19)$$

where

$$\nu_c = -\frac{\hat{k}_0^2 L^2}{4a_2 a_{00}} [q_{00} + F(\omega)(q_{00} - a_{00})], \quad (20)$$

$$\mu_c = -\frac{\hat{k}_0^2 L^2}{4a_2 a_{00}} \left[F(\omega) \left(a_{00} - \frac{q_{00} q_2}{a_2} \right) - \frac{q_{00} q_2}{a_2} \right]. \quad (21)$$

The equation (19) represents the confluent Heun equation that can be solved analytically, see Appendix B.

Comparing Eq. (19) with Eq. (B.1) and using the relations (B.3) we get that

$$\alpha_c = 0, \quad \beta_c = -\frac{1}{2}, \quad \gamma_c = 0, \quad \delta_c = \mu_c, \quad \eta_c = \nu_c + \frac{1}{4}. \quad (22)$$

If we substitute these parameters into the general solution of the confluent Heun equation (B.6), then we get the general solution of Eq. (19) as

$$P(\xi) = AHc \left(0, -\frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; \xi \right) + B\xi^{\frac{1}{2}} Hc \left(0, \frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; \xi \right), \quad (23)$$

where A and B are integration constants and $Hc(\alpha_c, \beta_c, \gamma_c, \delta_c, \eta_c; \xi)$ stands for the confluent Heun function.

Returning to the original variable x , we can write the general solution of Eq. (15) as

$$\begin{aligned} P(x) &= AHc \left(0, -\frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; a_2 \left(\frac{x}{L} - 1 \right)^2 \right) + B \left(\frac{x}{L} - 1 \right) Hc \left(0, \frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; a_2 \left(\frac{x}{L} - 1 \right)^2 \right) \\ &\equiv AH_1(x) + B \left(\frac{x}{L} - 1 \right) H_2(x), \end{aligned} \quad (24)$$

where

$$H_1(x) \equiv Hc \left(0, -\frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; a_2 \left(\frac{x}{L} - 1 \right)^2 \right)$$

and

$$H_2(x) \equiv Hc \left(0, \frac{1}{2}, 0, \delta_c, \nu_c + \frac{1}{4}; a_2 \left(\frac{x}{L} - 1 \right)^2 \right).$$

It is worth noting that evaluating confluent Heun functions and their derivatives can be easily accomplished using mathematical software such as Maple (version 10 and later) and Mathematica (version 12 and later).

Furthermore, it is essential to mention that when $q_{00} = a_{00}$, the function $Y_w(x, \omega)$ is equal to zero. Consequently, the model equation (2) transforms into the classical Webster equation. In this scenario, the general solution (24) is an exact analytical solution for the Webster equation with the given variable cross-section.

5. Wave splitting and the calculation of the reflection coefficient

We will utilize the wave splitting method in the frequency domain (refer to [16]) to decompose the total acoustic field expressed in equation (24) into forward p_+ and backward p_- traveling waves. The wave splitting method is explained in detail in [17], and we will provide an overview of its use here.

Taking into account thermoviscous losses through the complex velocity, we utilize equation (A.2) to accomplish this task (for additional information, refer to Appendix A):

$$\frac{1}{\rho_0 c_0^2} \frac{\partial p'(x, t)}{\partial t} + \frac{v_y(x, t)}{a_0(x)} + \frac{v_x(x, t)}{a_0(x)} \frac{da_0(x)}{dx} + \frac{\partial v_x(x, t)}{\partial x} = 0, \quad (25)$$

where t is time, p' is an acoustic pressure, v_x is the component of an acoustic velocity in the x -axis direction and v_y is the component of an acoustic velocity in the y -axis direction.

Considering the harmonic acoustic field

$$p'(x, t) = \text{Re} [P(x) \exp(-j\omega t)], \quad v_x(x, t) = \text{Re} [V_x(x) \exp(-j\omega t)], \quad v_y(x, t) = \text{Re} [V_y(x) \exp(-j\omega t)] \quad (26)$$

and employing the relation (A.14), i.e. $V_y(x) = Y_w(x, \omega)P(x)$, we can rewrite Eq. (25) to account for losses as

$$\frac{dU(x)}{dx} = \left(\frac{j\omega a_0(x)}{\rho_0 \hat{c}_0^2} - Y_w(x, \omega) \right) P(x), \quad (27)$$

where $U(x) = a_0(x)V_x(x)$ and $Y_w(x, \omega)$ is the wall admittance.

We can express the linearized Euler equation for harmonic plane waves

$$\frac{dP(x)}{dx} = \frac{j\omega \rho_0}{a_0(x)} U(x) \quad (28)$$

and Eq. (27) in the matrix form

$$\frac{d}{dx} \begin{pmatrix} P(x) \\ U(x) \end{pmatrix} = \begin{pmatrix} 0 & \frac{j\omega \rho_0}{a_0(x)} \\ \frac{j\omega a_0(x)}{\rho_0 \hat{c}_0^2} - Y_w(x, \omega) & 0 \end{pmatrix} \begin{pmatrix} P(x) \\ U(x) \end{pmatrix}. \quad (29)$$

The functions $P(x)$ and $U(x)$ are independent and constitute a complete base of functions for the system of Eqs. (27) and (28).

Diagonalization allows us to make the transition from the base of functions $P(x)$ and $U(x)$ to the base of functions $P_+(x)$ and $P_-(x)$ (see e.g., [17]). After normalization, the following relation holds between these bases:

$$\begin{pmatrix} P_+(x) \\ P_-(x) \end{pmatrix} = \mathbb{M} \begin{pmatrix} P(x) \\ U(x) \end{pmatrix}, \quad (30)$$

where

$$\mathbb{M} = \begin{pmatrix} 1 & \frac{Z_e(x, \omega)}{a_0(x)} \\ 1 & -\frac{Z_e(x, \omega)}{a_0(x)} \end{pmatrix}. \quad (31)$$

Employing Eq. (3) we obtain the relation for the effective impedance:

$$Z_e(x, \omega) = \frac{\rho_0 \hat{c}_0}{\sqrt{\frac{\tan[k_0(q(x) - a_0(x))]}{k_0 a_0(x)} + 1}}. \quad (32)$$

Note that $Z_e(x=0, \omega) = \rho_0 \hat{c}_0$ when $q(0) = a_0(0)$.

Based on Eq. (30) and the linearized Euler equation $V_x = (-j/\omega \rho_0) dP(x)/dx$ we can write for the forward and backward pressure waves that

$$P_+(x) = \frac{1}{2} \left(P(x) + Z_e(x, \omega) V_x(x) \right) = \frac{1}{2} \left(P(x) - \frac{jZ_e(x, \omega)}{\omega \rho_0} \frac{dP(x)}{dx} \right), \quad (33)$$

$$P_{-}(x) = \frac{1}{2} (P(x) - Z_e(x, \omega) V_x(x)) = \frac{1}{2} \left(P(x) + \frac{j Z_e(x, \omega)}{\omega \rho_0} \frac{dP(x)}{dx} \right). \quad (34)$$

By summing (33) and (34) we get the total acoustic pressure field

$$P(x) = P_{+}(x) + P_{-}(x). \quad (35)$$

Furthermore, the following is true

$$V_{x+}(x) = \frac{P_{+}(x)}{Z_e(x, \omega)}, \quad V_{x-}(x) = -\frac{P_{-}(x)}{Z_e(x, \omega)}. \quad (36)$$

Then we can write that

$$V_x(x) = V_{x+}(x) + V_{x-}(x). \quad (37)$$

If we use approximations for the tangent function (11), then the effective impedance (32) can be expressed as

$$Z_e(x, \omega) = \frac{\rho_0 \hat{c}_0}{\sqrt{\frac{(q(x) - a_0(x))(1 + F(\omega))}{a_0(x)} + 1}} = \frac{\rho_0 \hat{c}_0}{Q(x, \omega)}. \quad (38)$$

For $F(\omega) = 0$ (applicable for very low frequencies) this relation reduces into the following one

$$Z_e(x) = \frac{\rho_0 \hat{c}_0}{\sqrt{\frac{q(x) - a_0(x)}{a_0(x)} + 1}}. \quad (39)$$

Recall that for the approximation (11) the overall acoustic pressure field is given by the solution (24).

Based on Eqs. (35)–(37), we can write that

$$Z(x, \omega) = \frac{P(x)}{V_x(x)} = Z_e(x, \omega) \frac{1 + R_e(x, \omega)}{1 - R_e(x, \omega)}, \quad (40)$$

where

$$R_e(x, \omega) = \frac{P_{-}(x)}{P_{+}(x)}. \quad (41)$$

It is important to note that the relation (40) differs from the relation (6).

By referring to Eq. (40), it is also possible to express that

$$R_e(x, \omega) = \frac{Z(x, \omega) - Z_e(x, \omega)}{Z(x, \omega) + Z_e(x, \omega)} = \frac{1 - Z_e(x, \omega) Y(x, \omega)}{1 + Z_e(x, \omega) Y(x, \omega)}. \quad (42)$$

By substituting the numerical solution of the Riccati equation (7) into Eq. (42), we can calculate the reflection coefficient $R_e(x, \omega)$ numerically.

When we substitute the acoustic impedance obtained from relation (40) into the reflection coefficient relation (8), we can derive the following expression:

$$R(x=0, \omega) = \frac{(\hat{c}_0 + c_0) R_e(x=0, \omega) + \hat{c}_0 - c_0}{(\hat{c}_0 - c_0) R_e(x=0, \omega) + \hat{c}_0 + c_0} \quad (43)$$

that represents the relation between the reflection coefficients $R(x=0, \omega)$ and $R_e(x=0, \omega)$.

The reflection coefficient $R_e(x=x_0, \omega)$, where $x_0 \in [0, L]$, is calculated based on the relation (41) and represents the reflection coefficient inside the SBH at $x=x_0$. Additionally, it is also possible to interpret the reflection coefficient $R_e(x=x_0, \omega)$ as representing the reflection from a homogeneous medium, where the region $[0, x_0]$ is replaced by a homogeneous medium with the effective impedance $Z_e(x=x_0, \omega)$ in a waveguide with a rectangular cross section of $S(x=x_0) = 2a_0(x_0)b$ (for more information, refer to e.g., [14]).

Throughout this work, we will refer to the reflection coefficient as $R(x, \omega)$, which is introduced in Sec. 3.

6. Validation of the general solution (24) and demonstration that the assumed structure behaves as the SBH

The validity of the presented general solution (24) has been established by comparing the reflection coefficient at $x=0$ based on this solution with the numerical solution of the Riccati equation (7). The numerical solution of the Riccati equation has been obtained using the standard Runge-Kutta-Fehlberg method (RKF45).

The following parameters for air are used in all calculations: $c_0 = 345 \text{ m s}^{-1}$, $\rho_0 = 1.193 \text{ kg m}^{-3}$, $a = 0.05$ ($\hat{c}_0 = c_0(1 - ja)$).

The geometric parameters of the SBH used for all calculations below are: $A_0 = 0.02 \text{ m}$ ($f_{\text{off}} = 4312 \text{ Hz}$), $b = 2A_0$ (square cross-section), $a_{00} = 0.02$, $L = 0.15 \text{ m}$, $q_0 = 0$, $x_0 = 0$.

We assume that SBH is terminated by the perfectly rigid wall, i.e. $R(x=L, \omega) = 1$ ($Y(x=L, \omega) = 0$). This means that

$$\left(\frac{dP(x)}{dx} \right)_{x=L} = 0. \quad (44)$$

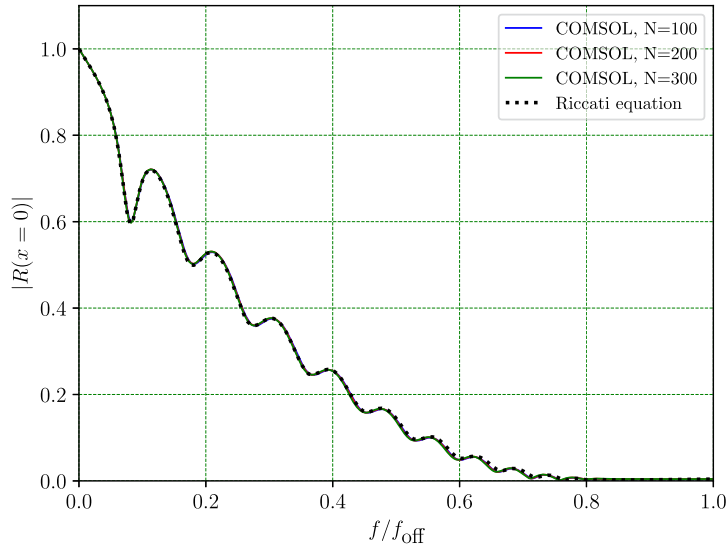


Fig. 4. Comparison of the numerical results: Reflection coefficient modulus calculated by the 2D FEM model for various number of ribs, and calculated employing 1D model based on the Riccati equation (7).

Based on the general solution (24) we can write that

$$\frac{dP(x)}{dx} = A \frac{2a_2 \left(\frac{x}{L} - 1 \right)}{L} H_1'(x) + B \frac{2a_2 \left(\frac{x}{L} - 1 \right)^2}{L} H_2'(x) + B \frac{H_2(x)}{L}. \quad (45)$$

It follows from the condition (44) that

$$\left(\frac{dP(x)}{dx} \right)_{x=L} = \frac{B}{L} = 0. \quad (46)$$

From here it follows that $B = 0$.

Note that

$$H_1(L) = H_2(L) = 1 \quad \text{and} \quad \left(\frac{dH_1(x)}{dx} \right)_{x=L} = \left(\frac{dH_2(x)}{dx} \right)_{x=L} = 0. \quad (47)$$

From the relation (6) it follows that

$$R(x, \omega) = \frac{Z(x) - \rho_0 c_0}{Z(x) + \rho_0 c_0} = \frac{\frac{P(x)}{V_x(x)} - \rho_0 c_0}{\frac{P(x)}{V_x(x)} + \rho_0 c_0} = \frac{P(x) + \frac{j}{k_0} \frac{dP(x)}{dx}}{P(x) - \frac{j}{k_0} \frac{dP(x)}{dx}}, \quad (48)$$

where based on the linearized Euler equation we put $V_x(x) = -j/(\omega \rho_0) dP(x)/dx$, $k_0 = \omega/c_0$.

If we consider that $B = 0$, then substituting (24) and (45) into (48) we obtain that

$$R(x, \omega) = \frac{Lk_0 H_1(x) + j2a_2 \left(\frac{x}{L} - 1 \right) H_1'(x)}{Lk_0 H_1(x) - j2a_2 \left(\frac{x}{L} - 1 \right) H_1'(x)}, \quad (49)$$

where $H_1'(x) \equiv dH_1(x)/dx$.

Regarding the equalities (47), it is evident that $R(x = L, \omega) = 1$.

Prior to delving into the applicability of the analytical solution that we have derived, it is essential to establish that the one-dimensional model equation (2) is capable of capturing the wave phenomena in the SBH.

To validate the numerical predictions of the one-dimensional mathematical model presented above, we compared its results with those obtained using a two-dimensional (2D) mathematical model that resolves the inner structure of the SBH. In the 2D model, the ribs are distributed along the x -axis, separated by a distance $w = L/N$, where N is the total number of ribs forming the SBH. The ribs are assumed to be infinitely thin. The length of each rib is set as $\ell_i = A - a_0(x_i) + \epsilon$, where $x_i = iw$ is the position of the i -th rib, and ϵ is a length correction factor accounting for end-effects at the interface between the individual slits and the main cavity.

The numerical results indicate that the value of the length correction factor resulting in the best agreement between the two models is $\epsilon = w/5$. Therefore, we can conclude that the one-dimensional model provides accurate results even when compared to the more detailed 2D model, and that the presented numerical predictions are reliable for various numbers of ribs, see Fig. 4. To model

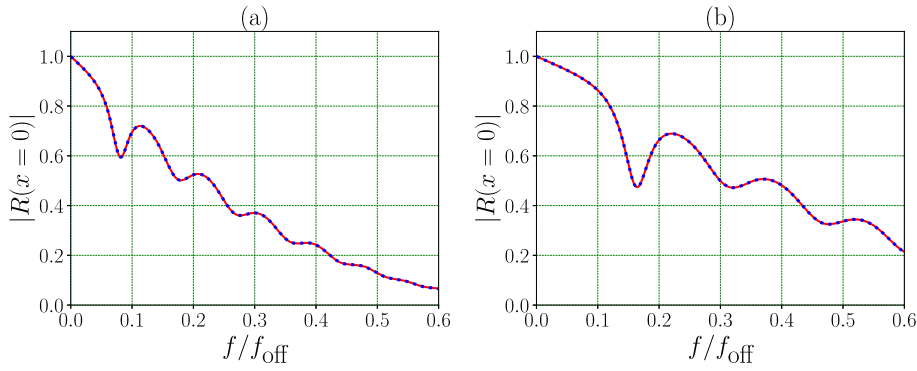


Fig. 5. Comparison of the dependence of the reflection coefficient at $x = 0$ on the frequency calculated from the relation (49) (solid red line) and the numerical solution of the Riccati equation (8) (dotted blue line) for the same the wall admittance (13): (a) $q_{00} = 1$ (b) $q_{00} = 0.2$.

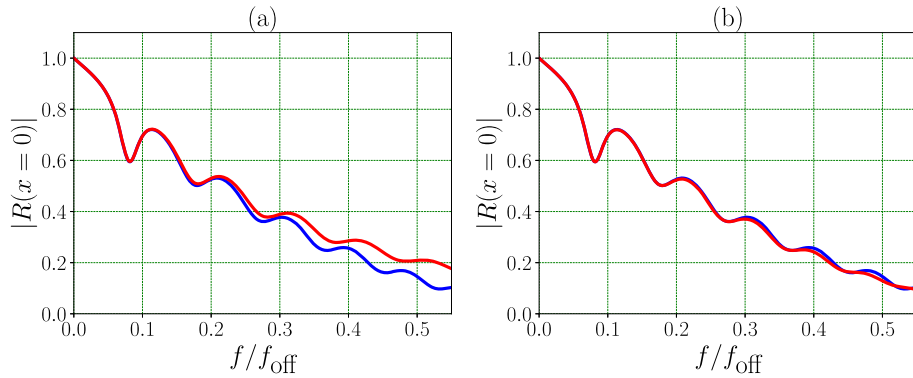


Fig. 6. Comparison of the frequency dependence of the reflection coefficient at $x = 0$ calculated from the relation (49) (red line) employing Eq. (13) and the numerical solution of the Riccati equation (8) (blue line) using Eq. (3); $q_{00} = 1$: (a) $F(\omega) = 0$ (the simplified approximation) (b) $F(\omega) \neq 0$.

the acoustic field in the SBH, we use the 2D Helmholtz equation:

$$\frac{\partial^2 P(x)}{\partial x^2} + \frac{\partial^2 P(x)}{\partial y^2} + \hat{k}_0^2 P(x) = 0. \quad (50)$$

We implement this 2D mathematical model using the finite-element (FEM) software COMSOL Multiphysics. Specifically, we use the Acoustics Module and the Pressure Acoustics Interface in the frequency domain. To account for the rigidity of the walls and ribs forming the SBH, we apply hard-wall boundary conditions at the fluid-solid interfaces. To discretize the computational domain, we use a free triangular mesh that is refined at the ends of the ribs. We ensure the numerical results are mesh-independent through careful testing.

Fig. 5 illustrates a comparison between $|R(x = 0, \omega)|$, obtained from the relation (49), and the numerical solution of the Riccati equation (7), for two cases: (i) the wall admittance (13), and $q_{00} = 1$ (constant wall function $q(x) = A_0$); and (ii) the wall admittance (13), and $q_{00} = 0.2$ (non-constant wall function $q(x)$).

As shown in the figure, the frequency dependencies of the absolute value of the reflection coefficient at $x = 0$ are identical, confirming the validity of the presented solution (24). Moreover, the effect of the wall function $q(x)$ on the frequency dependence of the reflection coefficient is evident.

Next, we aim to investigate the frequency range over which the approximate expression for the wall admittance (13) is valid. To accomplish this, we consider the wall admittance in the Riccati equation (7) to be the non-approximated relation (3) with $R(x = L, \omega) = 1$. We present the results of our analysis in Figs. 6(a)–8(a) for different values of the parameter q_{00} . We compare the results obtained using the non-approximated wall admittance with those based on the approximated wall admittance, where we assume $F(\omega) = 0$ by replacing the tangent function in the expression for the wall admittance (3) with its argument. Based on these comparisons, we find that the simplified approximation is only valid for low frequencies.

When we assume that the term $F(\omega) \neq 0$ in the approximate relation for the wall admittance (13) is given by the relation (12), we can observe that this approximation provides better results over a wider frequency range compared to the case where the wall admittance is calculated using the non-approximated relation (3), as illustrated in Figs. 6(b)–8(b). These findings suggest that the approximated wall admittance (13) can be utilized over a wider frequency range than the simplified approximation where $F(\omega) = 0$.

To demonstrate that the considered structure behaves as an SBH, we will calculate the phase and group velocity at $x = L$. We assume that the SBH terminates at a perfectly rigid wall, i.e., $R(x = L, \omega) = 1$. As previously mentioned, the integration constant B is

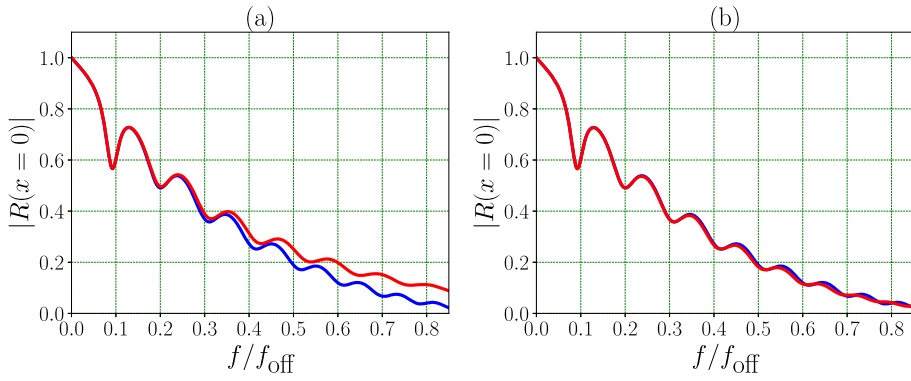


Fig. 7. Comparison of the frequency dependence of the reflection coefficient at $x=0$ calculated from the relation (49) (red line) employing Eq. (13) and the numerical solution of the Ricatti equation (8) (blue line) using Eq. (3); $q_{00} = 0.75$: (a) $F(\omega) = 0$ (the simplified approximation) (b) $F(\omega) \neq 0$.

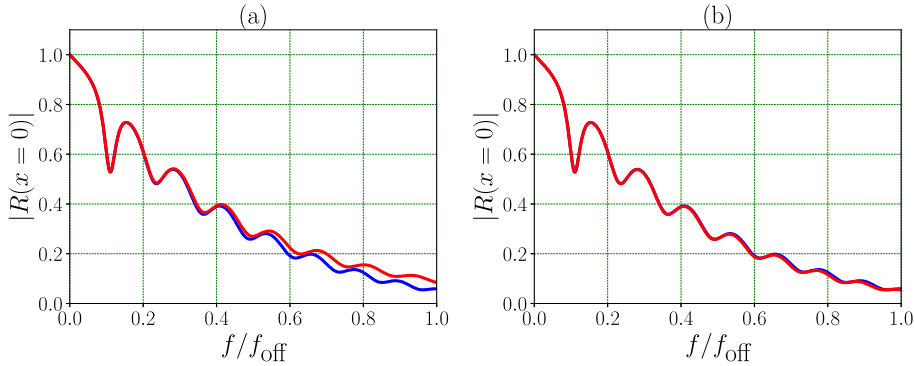


Fig. 8. Comparison of the frequency dependence of the reflection coefficient at $x=0$ calculated from the relation (49) (red line) employing Eq. (13) and the numerical solution of the Ricatti equation (8) (blue line) using Eq. (3); $q_{00} = 0.5$: (a) $F(\omega) = 0$ (the simplified approximation) (b) $F(\omega) \neq 0$.

zero. To demonstrate that no acoustic energy is lost and that the considered structure inherently behaves as a SBH, we assume $\hat{c}_0 = c_0$ ($a = 0$). We will consider the effective impedance $Z_e(x, \omega)$ in the form (38).

Under these conditions, the forward pressure wave (33) takes the following form:

$$P_+(x) = \frac{A}{2} \left[H_1(x) - \frac{j2a_2 Z_e(x, \omega) \left(\frac{x}{L} - 1 \right)}{\omega \rho_0 L} H_1'(x) \right]. \quad (51)$$

In order to find the phase velocity $c_{ph}(x, \omega)$ at the point $x = L$, we express the forward pressure wave as

$$P_+(x) = A_+(x) \exp(j\phi(x)), \quad (52)$$

where

$$\phi(x) = \arctan \left(\frac{-2a_2 \left(\frac{x}{L} - 1 \right) H_1'(x)}{k_0 L Q(x) H_1(x)} \right), \quad (53)$$

where (see the formula (38))

$$Q(x) = \sqrt{\frac{(q(x) - a_0(x))(1 + F(\omega))}{a_0(x)}} + 1. \quad (54)$$

From here we get that

$$k(x, \omega) = \frac{d\phi(x)}{dx} = -k_0 \frac{Q(x)P(x) \frac{d^2 P(x)}{dx^2} - P(x) \frac{dP(x)}{dx} \frac{dQ(x)}{dx} - \left(\frac{dP(x)}{dx} \right)^2 Q(x)}{k_0^2 Q^2(x) P^2(x) + \left(\frac{dP(x)}{dx} \right)^2}. \quad (55)$$

Using the formula (55) we can write the wavenumber at $x = L$ as

$$k(x=L, \omega) = \left(\frac{d\phi(x)}{dx} \right)_{x=L} = k_0 \sqrt{\frac{q_{00} + F(\omega)(q_{00} - a_{00})}{a_{00}}} . \quad (56)$$

For the phase velocity at $x = L$ we obtain

$$c_{ph}(x=L, \omega) = \frac{\omega}{k(x=L, \omega)} = \frac{c_0}{\sqrt{\frac{q_{00} + F(\omega)(q_{00} - a_{00})}{a_{00}}}} . \quad (57)$$

It is evident from relation (57) that the phase velocity decreases as the frequency increases.

In the limiting case, where the SBH is fully closed at its end, the phase velocity at the termination point can be expressed as follows:

$$c_{ph}(x=L, \omega) = \frac{c_0}{\lim_{a_{00} \rightarrow 0+} \sqrt{\frac{q_{00} + F(\omega)(q_{00} - a_{00})}{a_{00}}}} = \frac{c_0}{\infty} = 0 . \quad (58)$$

Using the relation (56) the group velocity at $x = L$ can be expressed as

$$c_g(x=L, \omega) = \left(\frac{dk(x=L, \omega)}{d\omega} \right)^{-1} = \frac{c_{ph}(x=L, \omega)}{1 + \frac{\omega(q_{00} - a_{00})c_{ph}^2(x=L, \omega)(dF/d\omega)}{2c_0^2 a_{00}}} . \quad (59)$$

For the limiting case we obtain

$$c_g(x=L, \omega) = \lim_{a_{00} \rightarrow 0+} \left(\frac{c_{ph}(x=L, \omega)}{1 + \frac{\omega(q_{00} - a_{00})c_{ph}^2(x=L, \omega)(dF/d\omega)}{2c_0^2 a_{00}}} \right) = 0 . \quad (60)$$

The obtained results in (58) and (60) demonstrate that the analyzed structure satisfies one of the fundamental properties of the SBH: the phase and group velocities become zero at the closing of the SBH, located at $x = L$ ($a_0(L) = 0$).

It should be noted that the considered structure exhibits sonic black hole behavior even when the wall function $q(x)$ ($q_{00} > a_{00}$) is taken into account.

In the special case where $q_{00} = a_{00}$, the model equation transitions into the Webster equation. This equation describes sound waves in a waveguide with variable cross-section and perfectly rigid walls. In this scenario, the structure no longer exhibits SBH attributes and $c_{ph}(x=L, \omega) = c_g(x=L, \omega) = c_0$.

7. Discussion

Through a comparison with the 2D mathematical model, we have demonstrated that the one-dimensional model equation (2) accurately captures the behavior of sound waves in the SBH, thereby confirming the relevance of the presented results based on the one-dimensional approach. Furthermore, these comparisons reveal that the one-dimensional model equation yields results that are in good agreement with the more accurate 2D model, even when different numbers of ribs are considered.

By comparing the analytical solution (24) of the model equation (2) with the numerical solution of the Riccati equation (8), we have demonstrated that the analytical solution for the given set of geometrical parameters remains valid up to approximately $f = 0.55f_{\text{off}}$ in the worst-case scenario ($q_{00} = 1$), when using the approximated relation for the wall admittance (13). Given that the practical application of the SBH is predominantly in the lower frequency range, the accuracy of the solution is deemed sufficient.

The accuracy of the solution is influenced by the parameter q_{00} in the wall function $q(x)$. Decreasing the value of this parameter improves the solution's accuracy for a given frequency, while increasing it has the opposite effect. In contrast, when $F(\omega) = 0$, the solution is only approximately valid up to $f = 0.23f_{\text{off}}$ in the worst-case scenario, which is half the frequency range that can be covered using the relation (12) in the approximated wall admittance ($F(\omega) \neq 0$).

While we acknowledge that the loss mechanisms cannot be accurately described by the complex propagation velocity $\hat{c} = c_0(1 + ja)$, incorporating losses more accurately falls beyond the scope of this paper. Our primary aim is to present a new form of a SBH, for which we have derived the model equation and its analytical solution for the quadratic cross-sectional function. By utilizing the analytical solution, we have proven that the sound-absorbing structure in question meets the fundamental requirements of the SBH. Specifically, we have confirmed that both the group and phase velocities are zero at the end of the structure, irrespective of frequency, when the cross-sectional function $a_0(x=L) = 0$.

8. Conclusion

We have introduced a new type of sound black hole (SBH) with a rectangular cross-section, which differs from previous research on SBHs with circular cross-sections. To investigate the behavior of acoustic waves in our proposed SBH, we derived a one-dimensional model equation and verified its applicability by comparing it with a 2D mathematical model. Our analytical solution for the one-dimensional model equation applies to a frequency domain where SBH properties are exhibited and considers

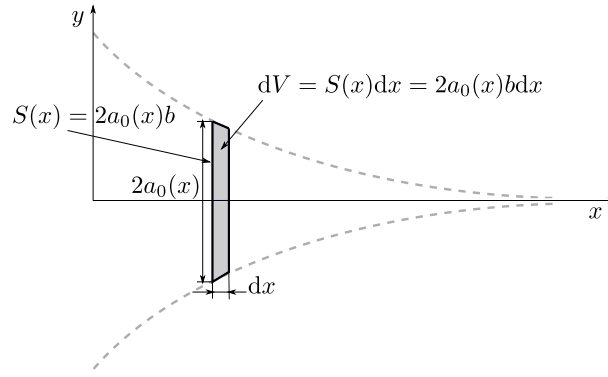


Fig. A.9. The volume element of SBH.

thermoviscous losses through the complex velocity. We also included a wall function in our solution to control the length of ribs. When the wall function is the same as the cross-sectional function, the model equation becomes the classical Webster equation, which has an exact analytical solution.

To test the accuracy of our solution, we derived the Riccati equation for the acoustic admittance, which can be solved numerically. Using the method of wave splitting in the frequency domain, we separated the overall acoustic field into forward and backward traveling waves, allowing us to not only calculate the reflection coefficient inside the SBH but also determine the phase and group velocity at the end of the SBH. Our calculations demonstrated that our proposed SBH satisfies the basic attributes of SBHs. We also discussed the applicability of our solution and anticipate that it will attract the attention of the scientific community. Further research could explore more realistic losses in SBHs or seek analytical solutions for other cross-sectional functions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Derivation of the model equation

To derive a model equation describing the acoustic field inside the SBH, we use the following one-dimensional continuity equation for the volume element (see Fig. A.9)

$$-\frac{\partial \rho(x, t)}{\partial t} dV = [-v_x(x, t)S(x) + v_x(x + dx)S(x + dx) + 2bv_y(x, t)]\rho. \quad (\text{A.1})$$

Here x is spatial coordinate in the direction of the SBH axis, t is time, $S(x)$ is an inner SBH cross-section, v_x is the component of the acoustic velocity in the x -axis direction, $v_y(x, t) = \partial \xi(x, t) / \partial t$, $\xi(x, t)$ is a displacement in the y -axis direction, $S(x) = 2a_0(x)b$ is the inner SBH cross-section, $dV = S(x)dx$ is the volume element and $\rho(x, t) = \rho'(x, t) + \rho_0$ is the total gas (air) density, where ρ' ($|\rho'| \ll \rho_0$) is the acoustic density and ρ_0 is the density corresponding to the equilibrium gas state.

Using Taylor expansion and subsequent linearization, we arrive at the following form of the continuity equation

$$\frac{1}{\rho_0 c_0^2} \frac{\partial p'(x, t)}{\partial t} + \frac{v_x(x, t)}{a_0(x)} \frac{da_0(x)}{dx} + \frac{v_y(x, t)}{a_0(x)} + \frac{\partial v_x(x, t)}{\partial x} = 0, \quad (\text{A.2})$$

where the acoustic density ρ' was replaced with the relation p'/c_0^2 , where p' is an acoustic pressure and c_0 is the sound velocity in the gas.

Using the linearized Euler equation

$$\frac{\partial v_x}{\partial t} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial x} \quad (\text{A.3})$$

and taking the derivative of the whole equation by time we obtain

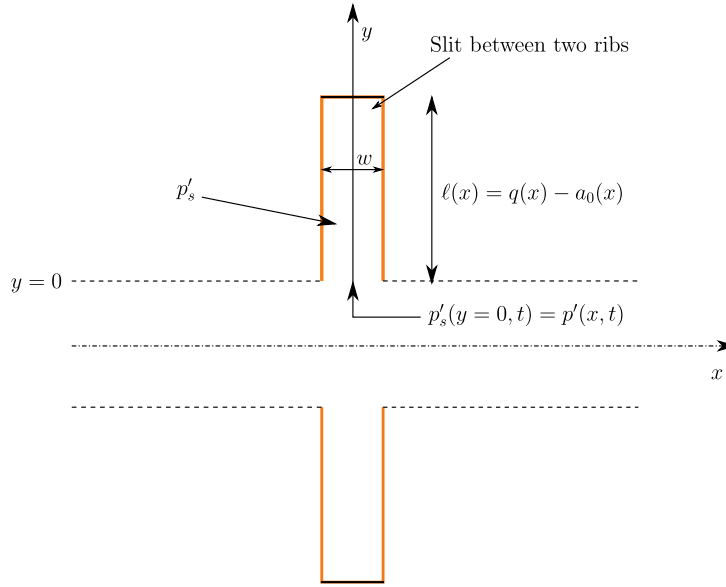


Fig. A.10. Situation picture for calculation of the acoustic field between two ribs.

$$\frac{\partial^2 p'}{\partial x^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{\partial p'}{\partial x} - \frac{\rho_0}{a_0(x)} \frac{\partial v_y(x, t)}{\partial t} = \frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2}. \quad (\text{A.4})$$

Since we consider the harmonic acoustic field, then

$$p'(x, t) = \text{Re} [P(x) \exp(-j\omega t)], \quad v_x(x, t) = \text{Re} [V_x(x) \exp(-j\omega t)], \quad v_y(x, t) = \text{Re} [V_y(x) \exp(-j\omega t)], \quad (\text{A.5})$$

where $j = \sqrt{-1}$ is the imaginary unit and $\omega = 2\pi f$ is the angular frequency, we can write Eq. (A.4) as

$$\frac{d^2 P}{dx^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{dP}{dx} + \frac{j\omega\rho_0}{a_0(x)} V_y(x) + k_0^2 P(x) = 0, \quad (\text{A.6})$$

where $k_0 = \omega/c_0$ is the wavenumber.

Next, we express $V_y(x)$ by investigating the acoustic field between two plates (ribs) ($w \ll b$), see Fig. A.10. The acoustic quantities used for the acoustic field in the slit between the ribs are given by the index s . The following wave equation is valid for the acoustic pressure $p'_s(y, t)$ in the slit:

$$\frac{\partial^2 p'_s}{\partial y^2} = \frac{1}{c_0^2} \frac{\partial^2 p'_s}{\partial t^2}. \quad (\text{A.7})$$

Let $p'_s(y, t) = P_s(y) \exp(-j\omega t)$, then this equation takes the form

$$\frac{d^2 P_s}{dy^2} + k_0^2 P_s = 0. \quad (\text{A.8})$$

Solving this equation for the following boundary conditions

$$\left(\frac{dP_s}{dy} \right)_{y=\ell} = 0, \quad P_s(y=0) = P(x), \quad (\text{A.9})$$

we obtain that

$$P_s(y) = P(x) \cos(k_0 y) + P(x) \tan(k_0 \ell) \sin(k_0 y). \quad (\text{A.10})$$

Using the linearized Euler equation, we obtain for the acoustic velocity inside the slit

$$V_s(y) = \frac{P(x)k_0}{j\omega\rho_0} (-\sin(k_0 y) + \tan(k_0 \ell) \cos(k_0 y)). \quad (\text{A.11})$$

From here we can write that

$$V_s(y=0) = -\frac{j k_0}{\omega \rho_0} \tan(k_0 \ell) P(x) = Y_w(x) P(x). \quad (\text{A.12})$$

Thus, the following relation holds for the wall admittance at the entry to the slit

$$Y_w(x) = -\frac{jk_0}{\omega\rho_0} \tan(k_0\ell(x)) = -\frac{jk_0}{\omega\rho_0} \tan(k_0(q(x) - a_0(x))). \quad (\text{A.13})$$

Since $V_s(y=0) = V_y(x)$, it is also valid that

$$V_y(x) = Y_w(x)P(x) \quad (\text{A.14})$$

We plug this relation into Eq. (A.6) to obtain the resulting model equation for the calculation of the acoustic pressure in the SBH:

$$\frac{d^2 P(x)}{dx^2} + \frac{1}{a_0(x)} \frac{da_0(x)}{dx} \frac{dP(x)}{dx} + \left(\frac{j\omega\rho_0}{a_0(x)} Y_w(x) + k_0^2 \right) P(x) = 0. \quad (\text{A.15})$$

It is worth noting that losses are not included in the model equation.

Appendix B. Confluent Heun equation and its solution

The canonical form of the confluent Heun equation (see e.g. [18,19]) is

$$\frac{d^2 U}{d\xi^2} + \left(\alpha_c + \frac{\beta_c + 1}{\xi} + \frac{\gamma_c + 1}{\xi - 1} \right) \frac{dU}{d\xi} + \frac{\mu_c \xi + \nu_c}{\xi(\xi - 1)} U(\xi) = 0, \quad (\text{B.1})$$

that has two regular singularities at $\{0, 1\}$ and one irregular singularity at ∞ .

The first solution of the confluent Heun equation in the vicinity of its regular singular point $\xi_0 = 0$ is possible to write by means of the Frobenius series (see e.g. [20,21])

$$Hc(\alpha_c, \beta_c, \gamma_c, \delta_c, \eta_c; \xi) \equiv U_1(\xi) = \sum_{n=0}^{\infty} c_n \xi^n, \quad |\xi| < 1, \quad (\text{B.2})$$

which is called the confluent Heun function. Here

$$\delta_c = \mu_c - \frac{\alpha_c(\beta_c + \gamma_c + 2)}{2}, \quad \eta_c = \nu_c - \frac{\beta_c}{2} - \frac{(\gamma_c - \alpha_c)(\beta_c + 1)}{2}. \quad (\text{B.3})$$

Substituting the series (B.2) into the confluent Heun equation (B.1) we obtain the following three-term recursion (see e.g. [18,19])

$$c_{n+1} = Q_n c_n + G_n c_{n-1}, \quad n \geq 0, \quad (\text{B.4})$$

where

$$Q_n = \frac{(n+1)^2 + (n+1)(\tau_c - 1) + \eta_c - \tau_c/2 - \beta_c(\alpha_c - \gamma_c)/2}{(n+1)(n+1+\beta_c)}, \quad G_n = \frac{\mu - \alpha_c(n+1)}{(n+1)(n+1+\beta_c)}, \quad (\text{B.5})$$

where $\tau_c = \beta_c + \gamma_c - \alpha_c$.

The initial conditions are $c_0 = 1$ and $c_{-1} = 0$.

The confluent Heun function $Hc(\xi)$ is analytic in the circle $|\xi| < 1$ and Cauchy's theorem on the expansion of an analytic function into a power series guarantees that the series (B.3) converges to $Hc(\xi)$ inside the circle $|\xi| < 1$, see e.g. [22].

Note that we can extend the local confluent Heun functions outside the circle of convergence of the series. The power series expansion at an arbitrary regular point is presented for example in [22]. When it comes to evaluating the confluent Heun functions, it is possible to use, for example, the mathematical software Maple (version 10 and higher) and Mathematica (version 12 and higher).

Employing the F -homotopic transformation (elementary power transformation), see e.g. [18,21], we obtain the second linearly independent solution of the confluent Heun equation (B.1). Then we can write the general solution of the confluent Heun equation in the vicinity of $\xi_0 = 0$ as

$$U(\xi) = AHc(\alpha_c, \beta_c, \gamma_c, \delta_c, \eta_c; \xi) + B\xi^{-\beta_c} Hc(\alpha_c, -\beta_c, \gamma_c, \delta_c, \eta_c; \xi), \quad (\text{B.6})$$

where A and B are integration constants.

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